- `Before-actor-system-terminate`—The tasks set here are executed after the cluster is shut down but before the system is shut down.

In this projection, the producer releases its registration before the `ActorSystem` terminates. This is shown in the following listing along with the rest of the producer's build.

Listing 15.5 Kafka producer creation

```
object MarketProjection {

  def init(system: ActorSystem[_]): Unit = {...}                    ❶

  def createProducer(
      system: ActorSystem[_]): SendProducer[String, Array[Byte]] = {

    val producerSettings =
      ProducerSettings(                                             ❷
        system,
        new StringSerializer,                                       ❸
        new ByteArraySerializer)                                    ❸
    val sendProducer = SendProducer(producerSettings)(system)
    CoordinatedShutdown(system).addTask(                            ❹
      CoordinatedShutdown.PhaseBeforeActorSystemTerminate,
      "closing send producer") { () =>
      sendProducer.close()
    }
    sendProducer
  }

  private def createProjection(
      system: ActorSystem[_],
      topic: String,
      producer: SendProducer[String, Array[Byte]],
      index: Int)
      : AtLeastOnceProjection[Offset, EventEnvelope[Market.Event]] = {...} ❺
}
```

❶ Initialization of the projection (ignored here)

❷ Loads parameters from the akka.kafka.producer configuration

❸ Configures the producer settings for serialization for the key and value